

Data Professional Survey Dashboard Report

1. Project Overview

The Data Professional Survey Dashboard is an interactive Power BI report developed to analyze responses from 630 data professionals across different countries, job roles, industries, salary ranges, and career experiences.

The objective of this dashboard is to provide insights into:

- Demographics of data professionals
- Salary distribution across job roles
- Popular programming languages
- Career transition trends
- Difficulty of entering the data field
- Employee satisfaction levels
- Work-life balance and salary happiness

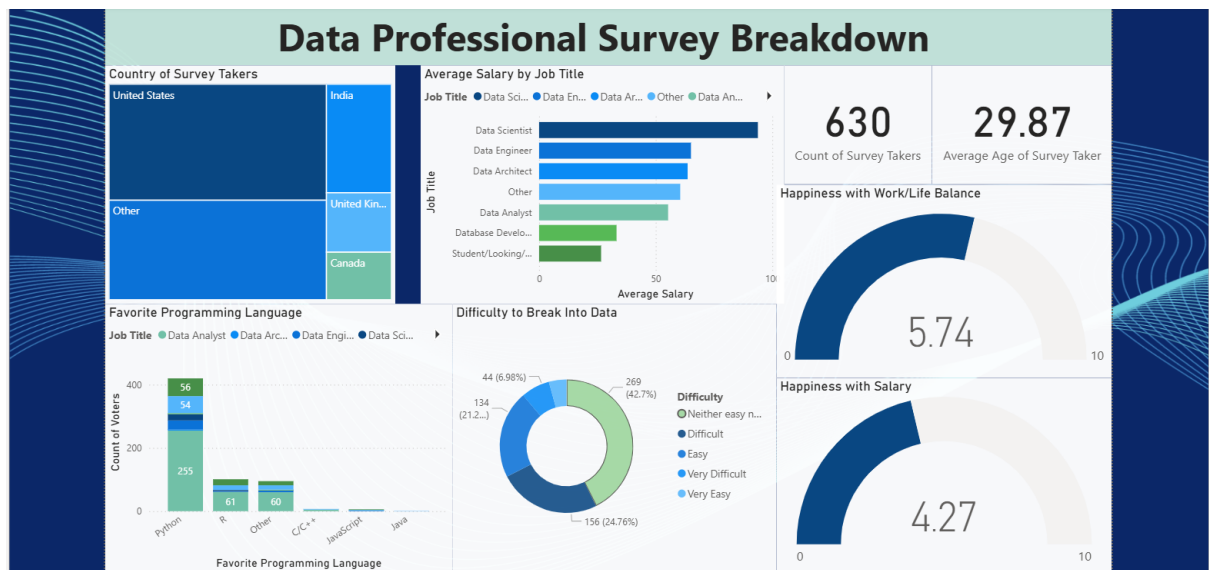
2. Dataset Description

- Dataset Name: Data Professional Survey
- Total Records: 630 Survey Responses
- Total Columns: 28 Columns

3. Data Cleaning and Preprocessing

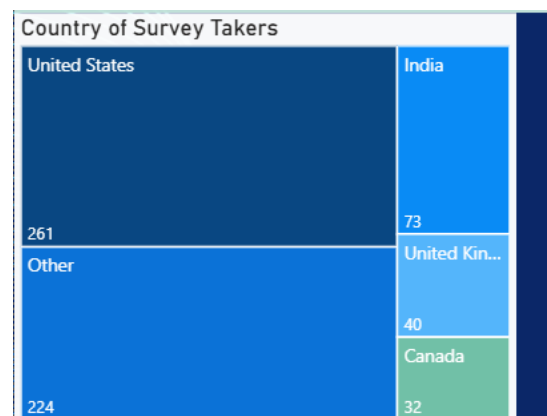
- Removed unnecessary columns- Browser, OS, City, Country
- Renamed Columns
- Handled missing values
- Standardised Job Titles (custom 'other' responses were grouped)
- Salary Transformation- Created an Average Salary column and then converted to numeric values.
- Data Type conversion
- Different new columns were created and unnecessary ones were removed

4. DASHBOARD:



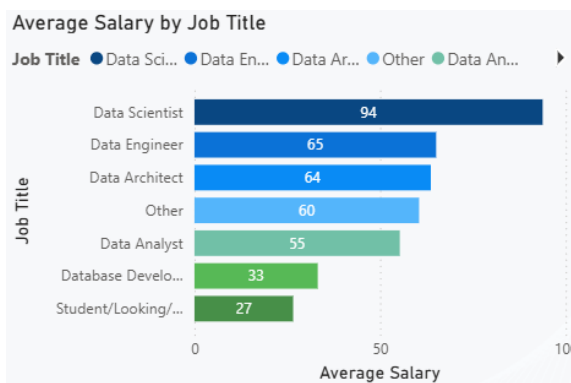
INSIGHTS:

Visual 1: Country of Survey takers



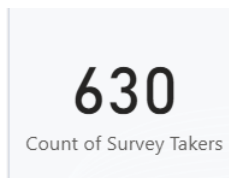
- **United States** has the highest number of respondents with **261 survey takers**, accounting for approximately **41.4%** of the total survey population.
- **India** is the **second-largest individual contributor** with **73 respondents (11.6%)**, indicating strong participation from the Indian data professional community.
- The **"Other Countries"** category contributes **224 respondents (35.6%)**, showing that the survey collected responses from a diverse range of countries worldwide.
- The **United Kingdom** contributed **40 respondents (6.3%)**, representing a moderate share of the survey participants.
- **Canada** has the lowest participation among the major displayed countries, with 32 respondents (5.1%).
- Since the **United States** contributes the largest share of responses, overall insights related to salary, job satisfaction, and career trends may be more influenced by the U.S. market.

Visual 2: Average Salary by Job Title



- **Data Scientists** earn the highest average salary at **94K**, making them the top-paid professionals among all surveyed job roles.
- **Data Engineers** rank second with an average salary of **65K**, reflecting the high demand for data infrastructure and pipeline development skills.
- **Data Architects** closely follow **Data Engineers** with an average salary of **64K**, indicating the value organizations place on data architecture expertise.
- Professionals categorized under **Other job roles** earn an average salary of **60K**, suggesting that several specialized data-related roles also offer competitive compensation.
- **Data Analysts** earn an average salary of **55K**, which is lower than Data Scientists, Engineers, and Architects but still represents a solid earning potential within the data industry.
- **Database Developers** have an average salary of **33K**, significantly lower than other technical data roles in the survey.

Visual 3: KPI Cards Insight



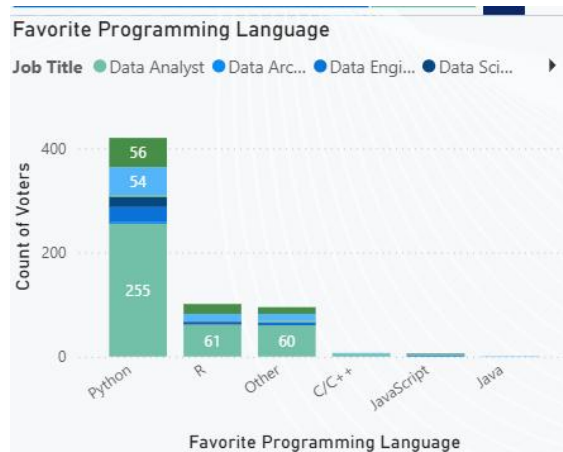
- A sample size of 630 respondents provides a strong foundation for understanding industry trends and making data-driven observations about the data profession.
- The sample size is large enough to identify meaningful trends in salary, job roles, programming language preferences, and job satisfaction.
- Responses were collected from professionals across multiple countries, making the analysis geographically diverse.



- The **average respondent age of 29.87 years** highlights that the data industry is predominantly driven by young and growing professionals.

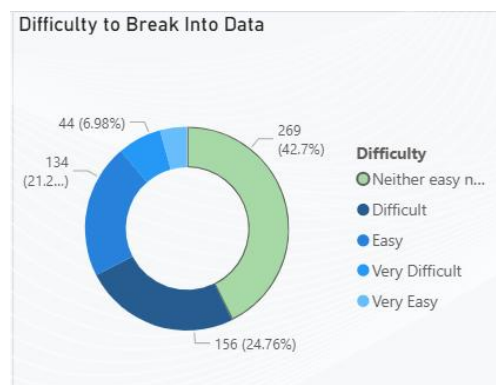
- Most respondents are likely in the **early to mid stages of their careers**, where skill development and career growth are key priorities.
- The relatively young workforce suggests a high level of adaptability to emerging technologies and evolving data tools.

Visual 4: Favourite Programming Language



- **Python** dominates all other programming languages with more than **400 votes**, making it the clear favourite among data professionals.
- Among Python users, **Data Analysts** contribute the **largest share (255 votes)**. This indicates that Python has become a standard tool for data cleaning, analysis, reporting, and business intelligence tasks.
- R receives significantly fewer votes than Python but remains popular among professionals involved in statistical analysis and research-oriented tasks.
- The "Other" category receives a notable number of responses, indicating that professionals also use specialized tools and languages depending on organizational requirements.
- C/C++, JavaScript and Java have limited Popularity.

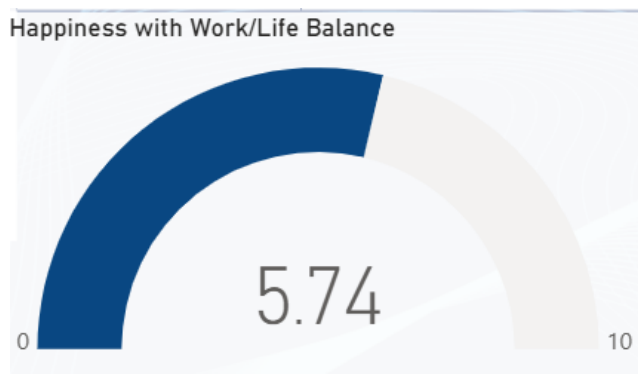
Visual 5: Difficulty To break Into Data



- 42.7% (269 respondents) believe entering the data field is neither easy nor difficult, making it the most common response.

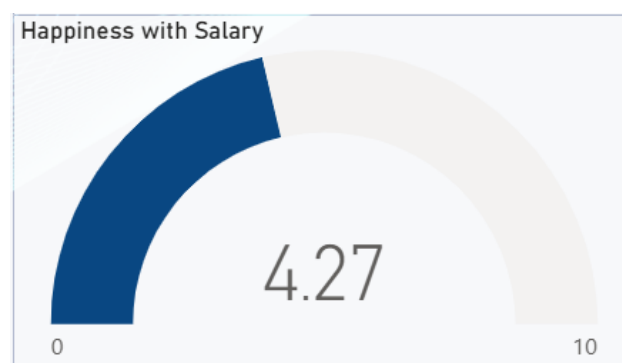
- 24.8% (156 respondents) find it difficult to break into the data industry.
- 21.3% (134 respondents) feel it is easy to enter the field.
- Only 7.0% (44 respondents) consider it very difficult.
- Very Easy received the lowest number of responses, showing that most professionals believe some effort and skills are required.
- Overall, the majority of respondents view the data industry as moderately accessible with proper training and experience.
- The results suggest that skills, projects, and practical experience play an important role in securing a data-related job.

Visual 6: Happiness with Work/Life Balance (5.74/10)



- The average work-life balance satisfaction score is 5.74 out of 10.
- Respondents show moderate satisfaction with their work-life balance.
- The score is slightly above the midpoint, indicating a generally balanced work environment.
- There is still room for improvement in employee well-being and flexibility.
- Better work-life balance initiatives could further increase employee satisfaction.

Visual 7: Happiness with Salary (4.27/10)



- The average salary satisfaction score is 4.27 out of 10.
- Respondents are less satisfied with their salaries compared to their work-life balance.
- The score is below the midpoint, indicating concerns regarding compensation.
- Salary appears to be one of the major areas of dissatisfaction among data professionals.
- Organizations may need to review compensation and reward structures to improve employee satisfaction.